Project Report

Group: 6 Thursday

Introduction

This report presents the results of a comprehensive surveying project undertaken as part of the Geoinformatics course. The primary objective of the project was to accurately map and document the spatial and physical features of a designated section of the campus, specifically focusing on the L7 Ground and the Western Lab Extension area. The surveyed region encompassed the L7 Ground along with its surrounding landscape elements, including the skywalk, staircases, internal roads, trees, and other nearby features. The project aimed to produce a precise and detailed representation of the area, contributing to a better understanding of its topography and infrastructure layout

Objective

The mapping exercise involved the following major tasks:

1. Conducting a **Reconnaissance Survey (Recy Survey)**
2. **Establishing a Control Network**
3. Performing **ETS Traversing**
4. Carrying out **Levelling using an Auto Level**
5. **Measuring Global Coordinates** of control points with **GNSS**
6. **Collecting offset data (features)** using a **Total Station** and storing it digitally
7. **Processing all collected data** to generate a detailed **topographic map** of the study area

The primary objectives of the survey camp for mapping the L7 Ground and Western Lab Extension area were as follows:

- To acquire hands-on experience in Total Station surveying and gain proficiency with key surveying instruments such as the Total Station, GNSS, and Auto Level.
- To prepare a detailed topographic map representing the L7 Ground and its surrounding landscape features.
- To apply GIS software, particularly **QGIS**, for processing field data and producing a comprehensive spatial map.
- To develop the ability to perform reconnaissance surveys, establish accurate control points, efficiently collect geospatial data, and ensure maximum area coverage with an optimal number of control stations.

Reconnaissance of Survey

The purpose of a reconnaissance survey is to collect preliminary information about a project area, done before detailed surveys and designs are carried out. It helps us to understand the general features, feasibility, and constraints of a site or region. A successful reconnaissance

ensures that the area to be surveyed is properly assessed, enabling the selection of optimal locations for control points. These serve as reference coordinates for mapping all features accurately.

Control Point Marking Guidelines:

- Should be in an open area with clear sky visibility (especially if ECEF coordinates are needed).

- Must cover the maximum possible area.

- Should be on a stable surface.

- Use the minimum number of control points necessary.

- Ensure intervisibility between consecutive control points

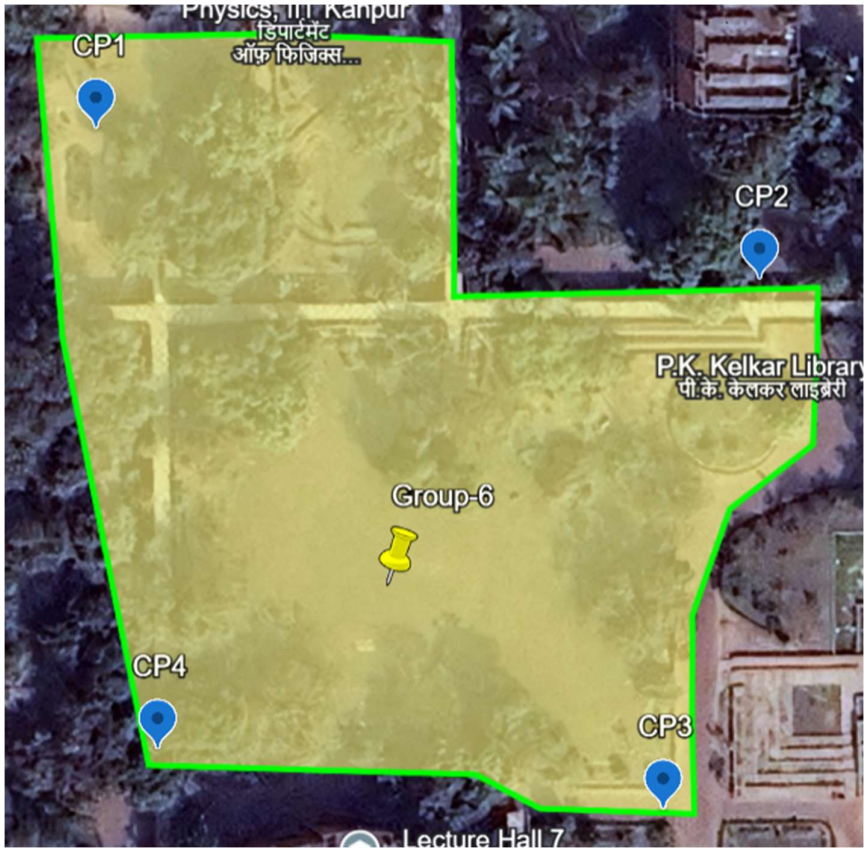


Figure 1:-Area allocated during groupwise distribution (https://earth.google.com/web/@26.51279226,80.23264455,159.08720386a,1000d,30y,0h,0t,0r/data=CgRCAggBOgMKATBCAggASggIr_ubpwYQAA)

Levelling

The basis purpose of this exercise is to to pass the height to the Control Point defined during the reconnaissance of Survey, from the point whose RL is already known to us.

Important Terms

- **Back Sight (B.S.)**: The back sight is a staff reading taken at a point of known elevation, such as a benchmark or change point. It is the first reading taken after setting up the instrument. Also, to establish the instrument height.
- **Fore Sight (F.S.)**: The fore sight is the second staff reading (after the back sight reading) taken on a point whose elevation is to be determined. It is used to establish the difference in elevation between two points.
- **Intermediate Sight (I.S.)**: An intermediate sight is any staff reading taken between the back sight and fore sight. It is used to determine the elevation of points between the instrument setup and the final station.
- **Change Point (C.P.)**: A change point is a point where the level instrument is shifted during a survey. Both back sights and fore sights are taken on this point to maintain a continuous line of measurement.
- **Bench Mark (BM)**: A benchmark is a fixed reference point of known elevation, typically used in surveying as a starting point for establishing elevations across a site or area.
- **Mean Sea Level (MSL)**: is the average height of sea surface calculated over a period of time.
- **Closing Error**: also known as misclosure, is the error in the reading of loop of points whose elevation is being calculated. It occurs when initial known elevation (some BM) doesn't match with the final calculated elevation (when we complete the loop and come back to initial point from where we started).
- **Datum Surface**: A datum surface is a precisely chosen reference level from which elevations are measured. In geodetic levelling we often chose Mean Sea Level (MSL) as a datum surface.
- **Reduced Level (RL)**: is the elevation of a point with respect to a known, fixed point or surface (Mean Sea Level in our case).

Methodology

(i) Setup and level the instrument
(ii) Take the first reading (back sight) by putting staff on the known RL (near GI lab)
(iii) Carryout fly levelling operation and calculate RL of some fixed features (Electric Poles and Markings on Ground) using two methods given above.
(iv) Take the last reading (foresight) and close on the same point near GI lab.

There are two methods used for Levelling:

1. Rise and Fall Method
   The arithmetic check applied for this method is
   $\sum B.S. - \sum F.S. = Last\ R.L. - First\ R.L$
2. Height of the instrument Method
   The arithmetic check applied for this method is
   $\sum B.S. - \sum F.S. = Last\ R.L. - First\ R.L$

For this project, we are doing Rise and Fall Method, to carry out all the leveling operations.

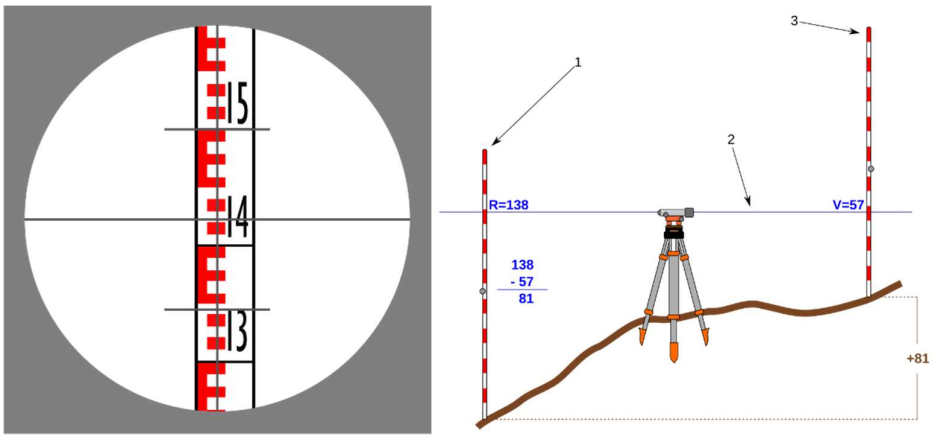


Figure 2:- Stadia marks on a crosshair and Diagram showing relationship between two level staff (https://en.wikipedia.org/wiki/Levelling)

Station	Point	Back Sight	Fore Sight	UL	LL	Rise	Fall	Reduced Level
	BM	1.13		1.217	1.006			128.409
			1.135	1.24	1.03		0.005	12.404
		1.011		1.267	0.761			
			1.379	1.631	1.109		0.368	128.036
		1.351		1.572	1.148			
	CP1 (A)		1.342	1.591	1.098	0.009		128.045
		1.398		1.764	1.039			
			1.341	1.684	0.987	0.057		128.102
		1.279		1.558	1.011			
	CP2 (B)		1.388	1.633	1.164		0.109	127.993
		1.558		1.861	1.257			
			1.587	1.892	1.296		0.029	127.964
		1.391		1.591	1.193			
	CP3 (C)		1.392	1.583	1.208		0.001	127.963
		1.384		1.649	1.128			
			1.379	1.629	1.123	0.005		127.968
		1.401		1.629	1.179			
	CP4 (D)		1.431	1.643	1.209		0.03	127.938
		1.359		1.86	0.848			
			1.268	1.886	0.635	0.091		128.029
		1.381		1.737	1.032			
			1.301	1.659	0.948	0.08		128.109
		1.466		1.828	1.096			
	BM		1.164	1.574	0.762	0.302		128.411
SUM---->>>>		16.109	16.107			0.544	0.542	

∑B.S - ∑F.S.=	0.002m
∑Rise - ∑Fall =	0.002m

Table 1;- levelling calculation (self made)

Closing Error

Misclosure: 128.409 – 128.411 = .002m

= 2mm

Given target was to kept the closing error below 10mm.

Methodology for Traversing:

To establish a close traverse using digital theodolite and adjust the closing error.

1. The Total Station was set up over a control point (say point A) by carefully centering and leveling the instrument.
2. A reference plane was established by placing a target prism on the backward control point (say point L).
3. The horizontal (internal) angle and slope distance between the two control points were measured.
4. After recording the measurements, the Total Station was shifted to the next control point (say point B), and the same process was repeated until all control points—from A to L—were observed.
5. For distance measurements, two readings were taken for each line, and their average value was used to improve accuracy and reliability.
6. Apply the check: Sum of External Angles of a Polygon = (n + 2) * 180
7. If the results differ, make adjustments.
8. The correction value (ξ) was computed as:
   ξ=Required Sum−Total Observed Angles
9. Divide the ξ by the number of sides and distribute the same value to each of the external
10. The external angles of the polygon are adjusted.
11. Since the lengths were not adjusted initially, any overshoot or undershoot was later compensated by adjusting the length of each traverse arm accordingly.
12. Calculate Whole Circle Bearing (WCB) at each control point.
13. Calculate Latitude and Departure of each control point i.e.

$$Latitude\ \Delta E = dSin\varphi$$

$$Departure\ \Delta N = dCos\varphi$$

14. Apply correction to the latitude and departure using Bowditch Rule.

$$\Delta E = (total\ correction\ (\sum \Delta E)\ /perimeter\ of\ the\ traverse) * segment\ length$$

$$\Delta N = (total\ correction\ (\sum \Delta N)\ /perimeter\ of\ the\ traverse) * segment\ length$$

15. After getting corrected latitude and longitude, find the local coordinates of the control points

16. Assume initial coordinate at first point and adding the same in other control point's coordinates find the local coordinates of all the control points.

	Station	Face			Horizontal Angle				
IS	Observed	(L/R)	Readings	In Decimal	Angle(°)	Mean	Correction (	Corrected A	Mean(°)
	B		0° 0' 0"	0°					
	D		77° 4' 21"	77.073°					
	B'		359° 59' 47"	359.996°					
	Error	L	0° 0' 13"	0.0036°	77.073		-0.1306028	77.2036028	
	B		180° 0' 0"	180°					
	D		257° 4' 22"	257.072°					
	B'		180° 0' 36"	180.010°					
A	Error	R	0° 0' 36"	0.010°	77.072	77.0725°	0.00385341	77.0681466	77.1359
	C		0° 0' 0"	0°					
	A		89° 40' 59"	89.683°					
	C'		0°.0' 13"	0.0036°					
	Error	L	0° 0' 13"	0.0036°	89.683		-0.1519708	89.8349708	
	C		180° 0' 0"	180°					
	A		89° 40' 53"	89.681°					
	C'		179° 59' 39"	179.994°					
B	Error	R	0° 0' 58"	0.016°	90.319	90.001°	0.00451572	90.3144843	90.0747
	D		0° 0' 0"	0°					
	B		97° 03' 31"	97.058°					
	D'		0° 0' 23"	0.006°					
	Error	L	0° 0' 23"	0.006°	97.058		-0.164468	97.222468	
	D		180° 0' 0"	180°					
	B		277° 03' 49"	277.063°					
	D'		180° 0' 23"	180.006°					
C	Error	R	0° 0' 23"	0.006°	97.063	97.0605°	0.00485291	97.0581471	97.1403
	A		0° 0' 0"	0°					
	C		95° 34' 21"	95.572°					
	A'		0° 0' 17"	0.004°					
	Error	L	0° 0' 17"	0.004°	95.572		-0.1619499	95.7339499	
	A		180° 0' 0"	180°					
	C		275° 33' 51"	275.564°					
	A'		180° 00' 01"	180.003°					
D	Error	R	0° 0' 01"	0.003°	95.564	95.568°	0.00477796	95.559222	95.6466
					Sum of all int	359.391°			359.9975
					Sum of all int	360.018°			
					Actual Sum c	360°			
					Misclosure (L	-0.609			
					Misclosure (R	0.018			

Table 2 : Internal angle calculations(self made)

• Angle Misclosure (Left Face) = $\sum\theta i - (n-2)*180°$

$= 359.391° - 360° = -0.609°$

• Angle Misclosure (Left Face) = $\sum\theta i - (n-2)*180°$

$= 360.018° - 360° = 0.018°$

$= 64.8$ seconds

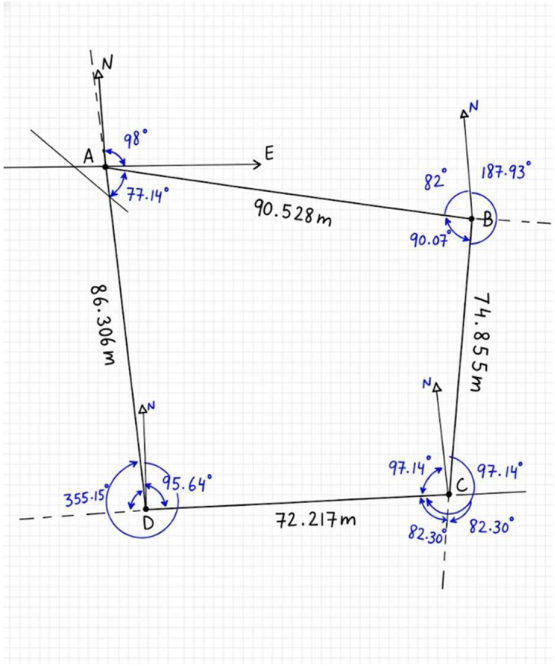


Figure 3: Bearing of all sides (self-made)

Line	WCB	Length(m)	Latitude			Departure		
			Calculated	Correction	Corrected	Calculated	Correction	Corrected
AB	98	90.528	-12.59906	0.071110257	-12.67017274	89.646988	-0.0523563	89.69934403
BC	187.93	74.855	-74.13919	0.058799027	-74.19798773	-10.327232	-0.043291919	-10.28394018
CD	270.79	72.217	0.9957037	0.056726863	0.938976884	-72.210135	-0.041766248	-72.16836921
DA	355.15	86.306	85.996978	0.067793852	85.92918381	-7.2969492	-0.049914533	-7.247034703
	Sum -->	323.906	0.2544302		2.22128E-07	-0.1873291		-6.4094E-08

POINTS	EASTING	NORTHING	ELEVATION
A	1000	1000	128.045
B	1089.7	987.3298273	127.993
C	1079.42	913.1318395	127.963
D	1007.25	914.0708164	127.938
A	1000	1000	

Table 3: Local Coordinates calculation(self made)

Feature Mapping

Feature mapping is the process of **identifying, recording, and representing natural and man-made features** on a map or plan of a survey area.

It's an important part of **topographical and engineering surveys**, providing detailed spatial information about the **physical characteristics of the site**.



Figure 4:- Total Station on a Tripod with a person holding prism (https://geospatial.trimble.com/en/products/hardware/trimble-s5)

A typical sequence of operations with total station involves the following stages:

1. Temporary adjustment and initialization of total station
2. Referencing/Targeting
3. Creation of Job file
4. Observations (Aiming and Measurement)
5. Transfer of data to computer
6. Use of QGIS to prepare the final map

Temporary adjustment and initialization of total station

1.Set up the total station by mounting it on the tripod and roughly leveling the setup.

2. Proceed with the leveling process as follows:

- Center the circular bubble using two leveling screws at a time, rotating both either inward or outward simultaneously.
- Rotate the instrument approximately 90 degrees from its previous position, and use the third leveling screw to center the bubble. This completes the leveling process.
- Repeat steps (a) and (b) if necessary.

3. Attach the batteries as per the instructions and press the POWER (PWR) button to start the instrument.

4. Perform fine leveling using the electronic level on the display, applying the principle of "three leveling screws."

5. Once fine leveling is done, press ENTER to initialize the instrument, which will switch to P0 mode (standard mode).

6. Enter the required values to initialize the total station, such as:

- Temperature, pressure, offsets, and relative humidity. After entering each value, the correction factor will be calculated and displayed.

7. Set the horizontal reference (HA Ref): Align the instrument approximately north, set the HA reference to 0° 0' 0", and press ENTER. The instrument will then display the horizontal angle (HA) and vertical angle (VA).

Referencing

- Press the MENU button.
- Press 3 to access the coordinate settings.
- Press 1 to go to station coordinate settings (coordinates of the total station).
- Enter the station number (stn) where the total station is set up (e.g., 1).
- For HT (height of the instrument), press ENTER to confirm. Measure the instrument height and press ENTER to input the value.

Next, establish a reference station or target. A reference station or target is a point where the coordinates (Northing, Easting, Elevation) are known, serving as the starting point for your work from the current station.

When prompted for:
- Northing (N), Easting (E), and Elevation (ELE): Enter the coordinates as a triplet (X, Y, Z), for example, 1000, 1000, 100.
- Instrument height (Ih): Enter the height of the instrument.

The system will then ask for the horizontal angle reference (HA Ref):
- You can either sight approximately north and press ENTER (0° 0' 0"),

(OR)

- If there is a reference point available, set up the reflector at that point and sight the reference target.
  - Press F6 and then ENTER to input the signal height (SH), typically 1.5 m.
  - Press the A/M (Aim/Measure) button. This will provide the coordinates of the reference station.

The instrument will switch to P0 mode and display the horizontal (HA) and vertical angles (VA).

Creation of Job File

A job file is where survey data is recorded. It is recommended to maintain one job file for each specific project. Multiple job files can be created using the same UDS to record different types of work. You can also recall an existing job file and continue adding data to it.

Procedure for Creating a JOB File in a Total Station:

1. Press the PRG button.
2. Enter the UDS program number (e.g., 5; typically ranging from 1 to 19) that you have created.
3. Enter the job file name (e.g., "Myjob").
4. Three options will be displayed: Imem, Xmem, and Serial. Turn on Imem by pressing 1 (pressing 1 acts as a toggle to switch between on and off). Press ENTER.

5. The instrument will then prompt for all variables that have a label type of 3 (duplicating). For example, this includes station number (stn), instrument height (Ih), and signal height (SH).

Observations (Aiming and Measurement)

a) After completing initialization, UDS creation, referencing, and job file setup, you can begin recording observations. To do this, place the reflector at the point of interest (any point where coordinates need to be recorded), sight the prism, and press the A/M button. The readings (such as coordinates) will be displayed on the screen. Press the REG button to save the data to memory.

b) Sight the next point of interest and continue recording points using the A/M button, followed by the REG button to store each set of readings.

c) For accurate contouring and to create a detailed map, the reflector should not be moved more than 5-foot paces or 2 meters between points. The distance may vary depending on the terrain, work area, surrounding features, and other factors. For highly uneven terrains, reduce the distance between readings for greater precision.

d) Collect readings for various features such as building corners, electric poles, trees, road edges, etc., along with the terrain. Use a proper naming convention for features, such as recording trees as T1, T2, T3, etc. This makes it easier to identify different features when preparing the map.

e) To end the operation, press the PRG button and enter 0 to return to the standard P0 mode. This will save your job file.

The data was later processed in QGIS to generate detailed maps and contour lines, demonstrating the Total Station's ability to map terrain and identify features like buildings and roads.

Combining field data with QGIS software improved visualization and analysis, highlighting the value of integrating technology for spatial analysis.

Conclusions

Throughout these five weeklong projects we worked on different aspects and concepts of geoinformatics. In which we implement all the knowledge that we have learned earlier. From recay survey to levelling to feature mapping one by one, we used all our previous lab work and knowledge to make this project.

In week one we have undergone through the site and availability of features, and their presence, followed by control point establishment.

In week two we transferred the height from known point (BM) to all the control points ensuring the closing error limit.

Followed by week three where we carried out traversing and in last two week we take all of the features covering every topographical details. This project helps us to know how to implement all of the concepts together one by one and make out desired output.

Map

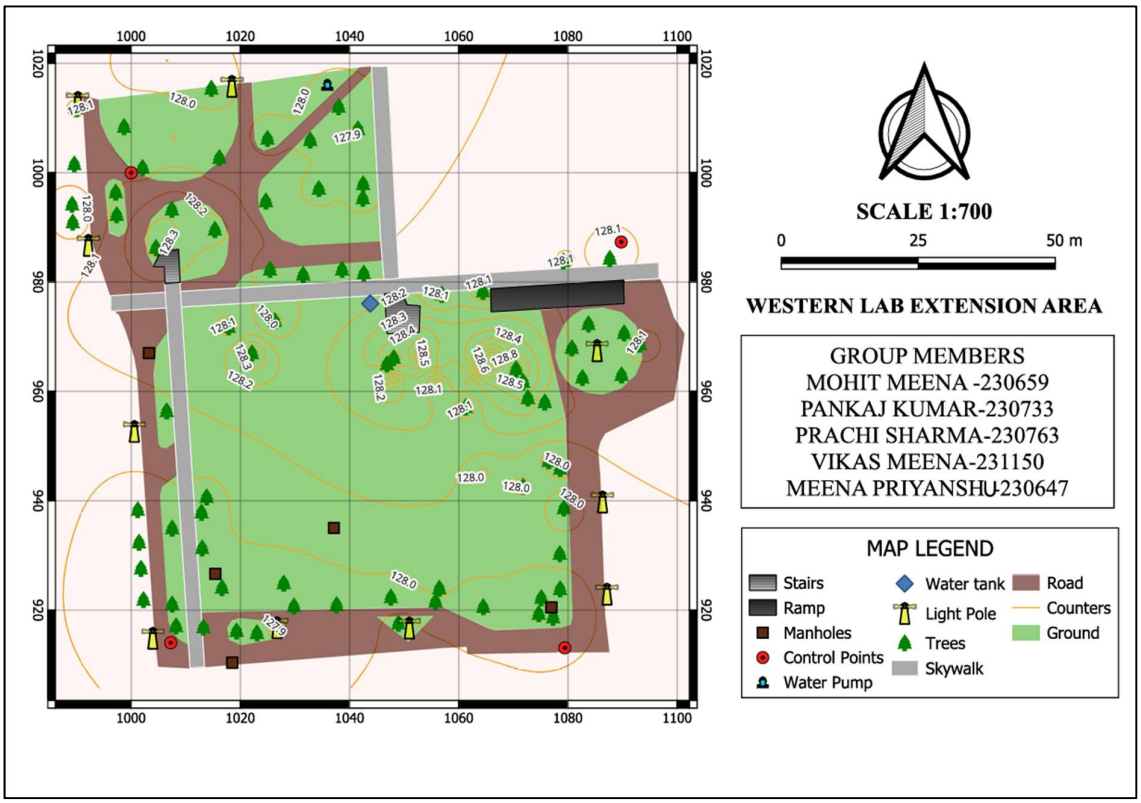


Figure 5:Overall Map (Self made by QGIS software)

References

All the references used in preparing this lab report are clearly mentioned.

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